

GRIFOLS Diagnostic Grifols, S.A.	Reference:	REGD-0000835	Version:	4.0, CURRENT
	Status:	Effective	Effective Date:	17-Aug-2011
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DG-57 Service Manual

“DISPENSE VERIFICATION” USER VERIFICATION PROCEDURE

Procedure description
Purpose: This dispensation control consists of checking the total level dispensed in each of the processed wells, detecting possible variations in this level due to potential problems caused by: • Drawing air due to the presence of bubbles in the sample tube or reagent vial. • Presence of air in the fluid system due to component (tubes, connectors) damage. • False sample level detection. This control is performed on tests that may cause false negatives due to an incorrect dispensation, i.e. tests which involve serum (or plasma) being dispensed or antiserum being added. In red blood cells tests, incorrect dispensation would be detected in the final reading, giving the result <i>empty</i> (<>) (insufficient red blood cells button.) Its purpose is to warn the user in case the detected liquid level doesn't fit software expectations. In any case, final decision of dispensation is under user's responsibility, as rejected cards are left in the service rack for inspection and results can be modified by user. Its purpose is not to provide a measure of the volume dispensed. Operation: The operation is as follows: 1. During cards identification step, the meniscus on top of supernatant liquid is detected and an initial volume value is stored per each well. 2. When the card is read after centrifugation, the meniscus on top of supernatant liquid is detected again and a final volume value is stored per each well. 3. The following calculation is done: • $\text{Dispensed volume} = \text{Final volume} - \text{Initial volume}$ Results and tolerances: The system will leave the card in the service rack for inspection in case that Dispensed volume differs from Expected volume by more than 12.5µl. This tolerance of 12.5µl is directly related to the variability of the calculation of the dispensed volume. The shape of the meniscus causes a variability of the detection height, which is the average height of several consecutive detections. Test files <i>.GRU</i> are available in Annex folder.

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EQUIPMENT

1. 8 test tubes of 15 x 100 mm filled with saline solution (0.9%)
2. 24 cards (Neutral type)
3. Manual pipette
4. Get *CHECKDISP_8DG.GRU* (DG Gel card) file.

SET UP

1. Turn Instrument on
2. Execute WADiana C. software
3. Wait until the instrument is initialized properly.

PROCEDURE

Process	Expected Results
Checking the correct behaviour 1. Place 8 test tubes in samples carousel. 2. Get 24 Cards place any card inside the incubator block, positions 1 to 24. 3. Execute <i>CHECKDISP_8DG.GRU</i>.	--- --- No volume dispensation error appears

Process	Expected Results
Simulating dispensation error 4. Place 8 test tubes in samples carousel. 5. Get 24 Cards place any card inside the incubator block, positions 1 to 24. 6. Execute <i>CHECKDISP_8DG.GRU</i>. 7. When the dispensation is finished, open the door and force the fault condition as shown below: 8. Aspirate 25 µL from each well of the cards number 1 to 4. 9. Dispense 25 µL to each well of the cards number 5 to 8. 10. Aspirate 25 µL from each well of the cards number 9 to 12. 11. Dispense 25 µL to each well of the cards number 13 to 16. 12. Aspirate 25 µL from each well of the cards number 17 to 20. 13. Dispense 25 µL to each well of the cards number 21 to 24. 14. Close the door. 15. Continue the process. 16. The result is showed in audit trail. In red blood cells tests, incorrect dispensation would be detected in the final reading, giving the result <i>empty (<=>)</i> (insufficient red blood cells button.)	--- --- --- --- --- --- --- --- --- --- --- --- --- --- Unexpected volume result should appear in all wells. Result ~ ~ ~ ~ ~

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DOCUMENT CHANGE HISTORY

Document change history			
Version	Date	Change	Rationale
2.0	2005/05/09	- Document REGD-0000835_v2 replaces document 57062V0M000. - Eliminated CHECKDISP list and added external reference to it, updated test list with DG-Gel one.	Uploaded document to GxPharma.
3.0	2011/04/28	- Eliminated in Step 4 references to CHECKDISP_8DN.GRU (DianaGel card), CHECKDISP_6DM (Diamed card). - Eliminated in Step 6: "CHECKDISP_NXX.GRU (N is a number and XX are letters depends from card used, for more details see equipment section)"	Service manual intended for Grifols market.
3.0	2011/04/28	Updated the Document Change History: - Date format changed to YYYY/MM/DD - Dates changed to document version effective date.	Adoption of ISO 8601 date format to avoid ambiguity.
3.0	2011/04/28	ID-GELSTATION and PROVUE erased from the title.	Service manual intended for Grifols market.
4.0	See document header.	Document REGD-0000835_v4 replaces document REGD-0000835_v3.	Document version changed.
4.0	See document header.	Document title.	Commercial reference substituted by project reference.
4.0	See document header.	Document header.	Increase information to easy document search.